■ RailConnect 2.0 — Project Overview

EXECUTIVE BRIEFING
High-Speed Rail Multi-WAN Platform

1. Mission Objectives

RailConnect is an autonomous multi-link onboard network control platform for high-speed trains (160–350 km/h). It eliminates connectivity drops caused by rapid cell tower handovers, Doppler shifts, and tunnel shadow zones by combining a 4-link heterogeneous uplink matrix, predictive ML proactive failover, and dynamic carriage LAN traffic shaping.

2. System Architecture & Information Flow

HETEROGENEOUS WAN INPUTS	RAILCONNECT CORE EDGE ENGINE	ONBOARD TRAIN CONSIST LAN
• **5G Alpha:** C-Band / mmWave (75 Mbps) • **5G Beta:** Sub-6 GHz Coverage (88 Mbps) • **Starlink LEO:** Satellite (165 Mbps) • **Wi-Fi 6E:** Station Gigabit (850 Mbps)	• **10s ML Predictor:** Random Forest failover. • **FastAPI + WebSockets:** 1Hz telemetry. • **Sub-50ms Handover:** Zero dropped packets. • **SQLite Time-Series:** Full audit trail.	• **Critical Ops:** Signaling, VoIP (100% SLA) • **Train Systems:** CCTV & Passenger Info • **Passenger Wi-Fi:** Adaptive Throttling • **Coach APs:** Loco, C1 (Exec), C2-C3, Pantry

3. Core Technical Differentiators

1. ML Proactive Switching

Evaluates signal decay rates and hazard distances to switch links **10 seconds ahead of failure**, preventing session drops.

2. Dynamic LAN QoS Slicing

When high-speed cellular slows down, LAN throttles guest streaming to **strictly protect 100% of train signaling & driver radio**.

3. Zero-Interruption Tunnels

Blackout zones automatically trigger **Critical Failsafe** on carriage APs, re-expanding to Gigabit Wi-Fi at station platforms.

4. Operational Telemetry Data Request (What We Need & Return Deliverable)

We are inviting partner railway operators, transit agencies, and hardware OEMs to provide operational modem logs to validate and calibrate predictive algorithms across diverse geographic corridors:

- **Requested Datasets:** 1 to 3 past trip logs (CSV or router exports) containing Timestamp, GPS coordinates, Speed (km/h), and Cellular modem metrics (RSRP, RSRQ, SINR, Latency, Bandwidth, Cell ID).
- **Absolute Privacy Guarantee:** Strictly physical RF parameters are required. **Zero passenger PII, MAC addresses, or payload content is collected.** All records can be anonymized and protected under mutual NDA.
- **Mutual Value Returned:** Participating organizations receive a complimentary **Route RF Quality & Dead-Zone Heatmap Report**, proving projected latency reductions and failover efficiency gains for their fleet.

RailConnect Research & Development Team • Live Prototype: http://localhost:8000
Contact: **contact@railconnect.io** • GitHub: **github.com/ashuyadav/railconnect-dashboard** • Version 2.4